

Cover Letter for PLOS ONE

Manuscript ID: PONE-D-26-13579R1

Dear Dr. Sun and Editors of PLOS ONE,

Please find enclosed the revised manuscript, “Computational surveillance of Colombian public pharmaceutical procurement using public administrative data: a reproducible analysis of a closed 2020-2025 cohort,” for reconsideration by PLOS ONE.

This study describes and evaluates a reproducible computational infrastructure for surveillance of Colombian public pharmaceutical procurement using public administrative data from SECOP-II, INVIMA, and SISMED. The manuscript reports the integration of these sources into an open analytical workflow combining relational storage, statistical anomaly detection, and a public-facing observatory. Using a corrected closed analytical cohort of 161,830 pharmaceutical contracts from 2020 to 2025 (after contract-level exclusion of 441 veterinary/agricultural records from a 162,271-contract source cohort), I identified statistically atypical procurement patterns, documented marked market concentration, and generated auditable outputs from otherwise fragmented public data sources.

I believe the manuscript remains well suited to PLOS ONE because its principal contribution is methodological and reproducible rather than journal- or country-specific. The work presents a transparent computational framework that can be evaluated, reused, and adapted in other settings where public procurement data are available but underused for systematic surveillance.

In this revision I have addressed every point raised by Reviewers #1 and #2 in a separate Response to Reviewers file. No changes to the financial disclosure are required: the study was conducted without external funding. The manuscript uses exclusively public, non-personal administrative data. No clinical, laboratory, or identifiable patient-level data were used. Under Colombian regulations, the study is classified as research without risk. The blockchain-based payment-cycle component is presented only as a complementary functional proof of concept and not as the central empirical contribution of the manuscript.

This submission is original, has not been published previously, and is not under consideration elsewhere. I have reviewed the manuscript, approve its submission, and agree to be accountable for the work. I have no competing interests to declare.

For reproducibility, the article-specific package is archived at Zenodo under DOI <https://doi.org/10.5281/zenodo.19074137> and is also available in the dedicated public repository at <https://github.com/zswamtech/BigLoI-PLOS-ONE-paper>.

Thank you for your consideration.

Sincerely,

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